Weak generalization on constants

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub thm:Generalization (this batch); pairing note: StrongGen and ChangeBdVar build on this card's construction.

Editorial note. The label thm:WeakGen names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Generalization theorem. Same unit, \ollabel{thm:Generalization} (source line 1310, this batch). Final step: from Gamma' proves subst(A,c,x) with x not free in Gamma' to the universal closure.

Source reference. Original theorem thm-Generalization: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 1309–1309. Prerequisite pointer invoked at proof line(s) 1345; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *(*Weak Generalization on Constants*) If* $\Gamma \vdash A$ *and* c *is constant not in occurring in* Γ*, then there is a variable* x *not in* A *such that* $\Gamma \vdash \forall x\, A[x/c]$*, and the proof does not involve* c*.*

Proof. Let $A_1, \ldots, A_n$ be a proof of A from Γ, so that $A = A_n$. Pick a variable x not in $A_1, \ldots, A_n$, and consider the new sequence: $A_1[x/c], \ldots, A_n[x/c]$. Such a sequence is a proof of $A[x/c]$ from Γ. In fact, for each $i = 1, \ldots, n$:

- if A is an axiom, then so is $A_i[x/c]$;
- if $A_i \in \Gamma$, then since c is not in Γ, we have $A_i[x/c] = A_i \in \Gamma$;
- if A_i is obtained from A_j and $A_j \to A_i$ then $A_i[x/c]$ follows by MP from $A_j[x/c]$ and $(A_j \to A_i)[x/c] = A_j[x/c] \to A_i[x/c]$.

It is clear that the constant c no longer occurs in the new sequence. Now let Γ' comprise those formulas from Γ that appear in the derivation of $A[x/c]$; then x is not free in any formula in Γ' and $\Gamma' \vdash A[x/c]$. By generalization (thm-Generalization) $\Gamma' \vdash \forall x\, A[x/c]$, whence by monotonicity $\Gamma \vdash \forall x\, A[x/c]$, as desired. $\square$